

Question	Marking Guidance	Mark	Comments
03.1	As size increases, ratio (of surface area to volume) decreases;	1	Accept converse. Comparison required, e.g., <u>smaller</u> organisms have a <u>larger</u> ratio
03.2	Two marks for correct answer in range of 1.75 to 1.76032;;	2	Accept for 1 mark, incorrect answer using radius 0.87 / 0.88 / 0.880 / 0.8802 / 0.88015; OR Accept for 1 mark, incorrect answer with correct rearranged equation, e.g., Radius = $\sqrt{(\text{surface area} \div 4\pi)}$ OR $= \sqrt{9.73 \div 12.56}$ OR $= \sqrt{0.77} / \sqrt{0.774} / \sqrt{0.775}$ OR $r^2 = \text{surface area} \div 4 \pi$ OR $r^2 = 9.73 \div 12.56$ OR $r^2 = 0.77 / 0.774 / 0.775$
03.3	(Measures) small uptake / amount / quantity / volume / concentration / rate (of oxygen uptake); OR Avoids use of powers of ten / standard form / many decimal places;	1	Ignore weight / accuracy

03.4	More accurate / less error (in measuring mass); OR Causes less distress / damage to animal (to measure mass); OR Easier / quicker (to find mass) because irregular shapes; OR Fewer measurements / calculations;	1	Ignore references to human error Accept converse if reference made to volume Reject if comparison is made with surface area.
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03.5	(Oxygen used in) respiration, which provides energy / ATP; OR (Oxygen is used in) respiration, which is a metabolic process / chemical reaction;	1	Reject produces energy Reject references to anaerobic respiration

03.6	- No information about egg; - So cannot compare all stages (in Table 2); - No statistical information / test / t-test / comparison of standard deviations; OR No measure of significant differences;	3	- Idea of comparing all three stages needed - Reject statements that "results" are not significant - Reject references to chi squared or correlation coefficient
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